



SEVERE WEATHER EXPOSURE FOR CRITICAL POWER INFRASTRUCTURE

Bernard Lawes

AGENDA

From customer problem to operational workflow

- | | | |
|----|---------------------------------|---|
| 01 | Why this problem matters | Grid resilience and the customer question |
| 02 | Spatial analysis | NOAA warnings × Overture power infrastructure |
| 03 | Results & scale | What the POC showed and what happens as workload grows |
| 04 | Operationalization | Parameterized job, managed output, and agentic tooling |
| 05 | Decision support | Interactive explorer, next steps, and customer discussion |

~15 MIN WALKTHROUGH • Q&A / DISCUSSION

WHY THIS PROBLEM MATTERS

Grid resilience is increasingly a geospatial problem.

01

Rising demand

More load is being placed on critical infrastructure while reliability expectations remain high.

02

Distributed threats

Severe weather, wildfire, equipment failure and deliberate disruption all have a geographic footprint.

03

Infrastructure can become the hazard

Damaged electrical infrastructure can amplify consequences beyond the asset itself, including wildfire ignition risk.

04

Spatial context connects the signals

Weather, vegetation, terrain, infrastructure condition, population and active incidents can be analyzed together.

THIS POC ESTABLISHES THE FIRST LAYER

Exposure

Risk

Prediction

Response

Severe Weather Exposure for Critical Power Infrastructure

Which critical power assets were exposed to severe thunderstorm and tornado warnings - and what warning context can help utility operations teams prioritize response?

EXPOSURE, NOT DAMAGE PREDICTION

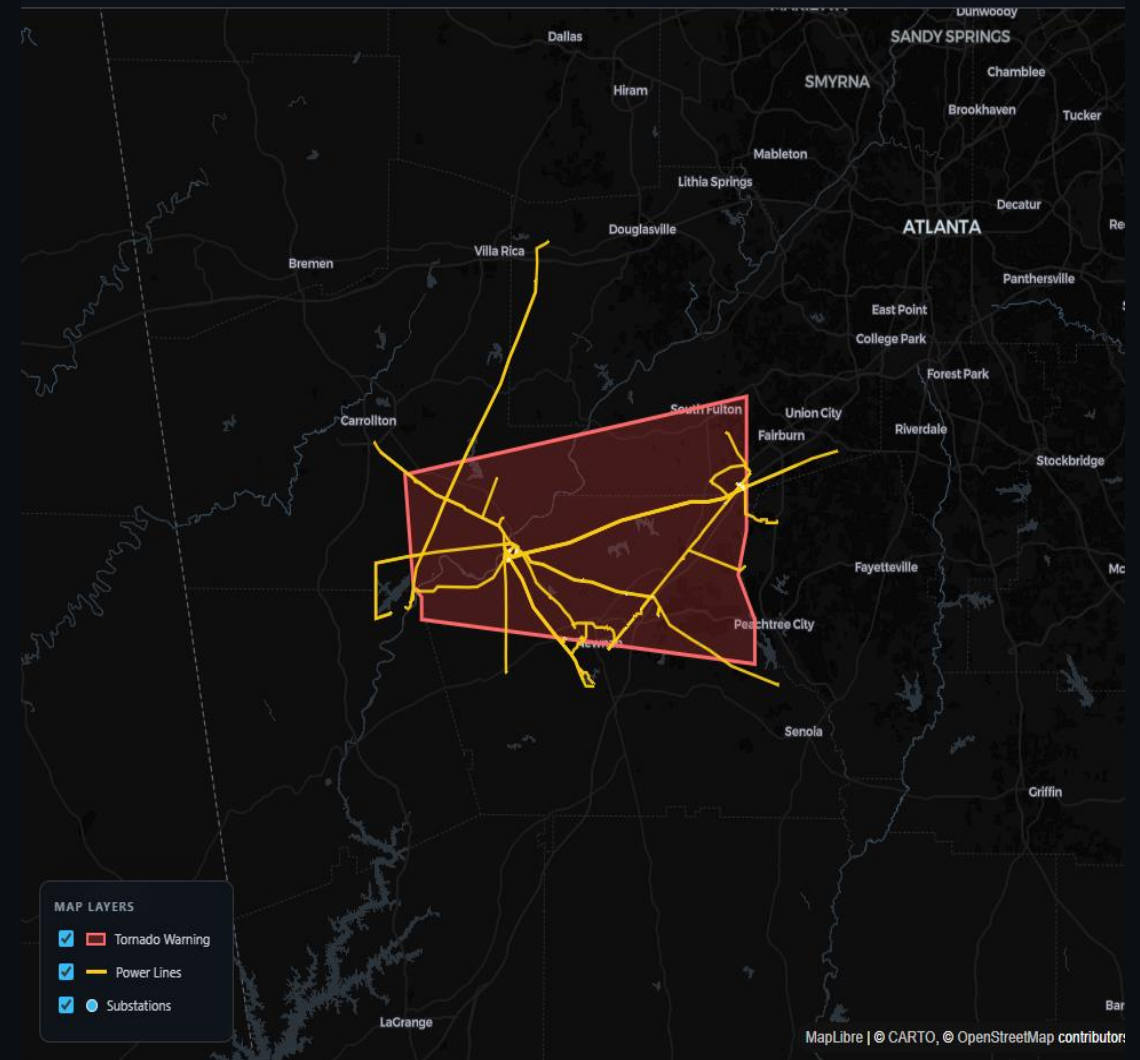
POC SUCCESS CRITERIA

- ✓ Identify exposed infrastructure
- ✓ Preserve warning characteristics
- ✓ Parameterize geography, year & asset classes
- ✓ Produce downstream-ready output

WHEREBOTS PROOF OF CONCEPT

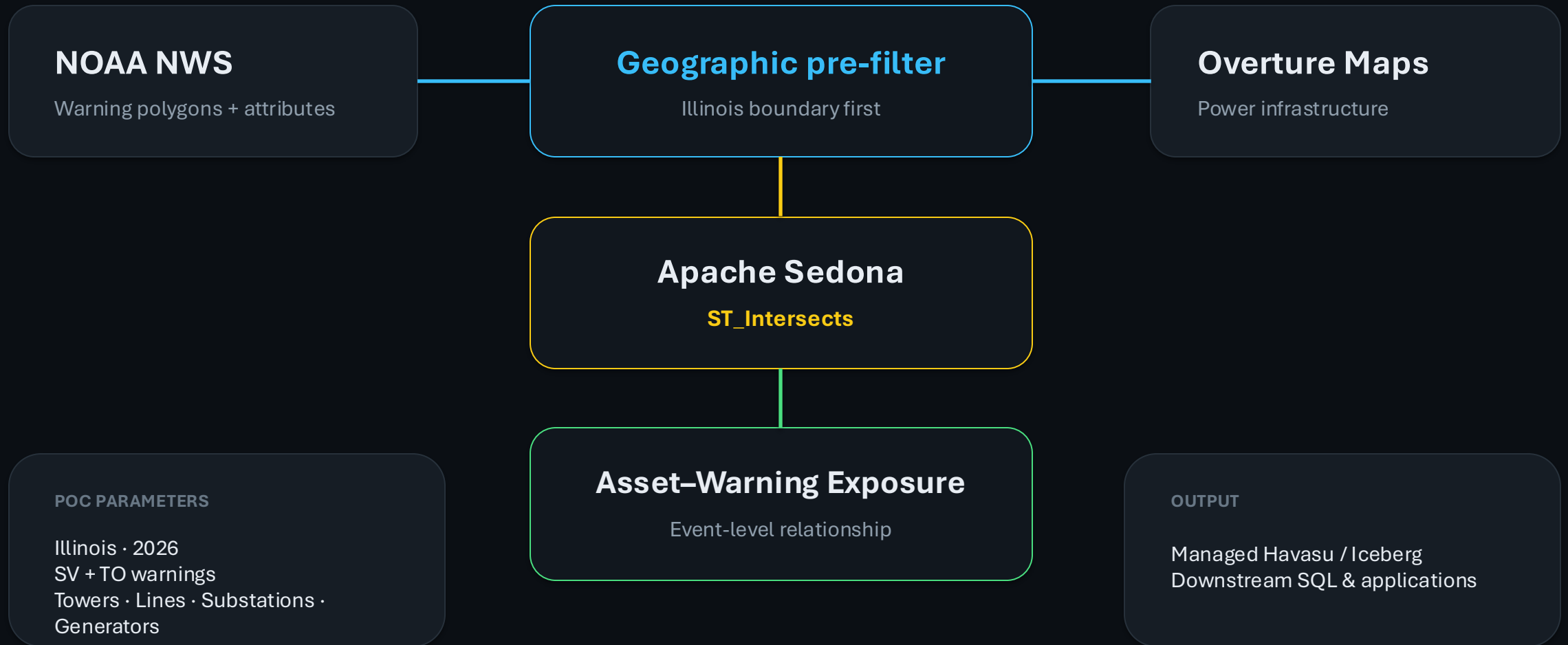
Severe Weather Infrastructure Exposure Explorer

Power infrastructure exposed to an observed tornado warning near Atlanta, Georgia.



From customer question to spatial analysis

Two datasets. One spatial relationship.



Illinois POC results

Exposure is widespread; warning characteristics add context.

1,572

NWS warnings

~134K

selected power assets

25 of 27

observed tornado warnings
intersected selected infrastructure

WARNING ACTIVITY

1,196 Severe Thunderstorm

376 Tornado

Wind tags: 50–100 mph • Hail tags: 0–4 in

KEY SA DECISION

Do not invent a risk score.

Exposure identifies the affected population. Customer-specific asset criticality, redundancy, outage history and operational context should drive prioritization.

Operationalizing the notebook

From interactive discovery to a repeatable Wherobots job.



WHY PYTHON SDK?

- Direct path from local development / CI
- No extra orchestrator required for the POC
- Explicit, inspectable, reproducible execution

OUTPUT SEMANTICS

createOrReplace

Deterministic POC reruns without duplicate exposure records.

Production history → run IDs · partitions · snapshots · MERGE

Scale exploration on a fixed Small runtime

8.1M exposure records without resizing the runtime.

Geography	Regions	Warnings	Assets	Exposures	Time
Illinois	1	1,572	134K	1.87M	194s
Midwest	5	5,412	444K	5.37M	209s
Midwest	10	8,134	852K	8.11M	101s
East Coast	15	4,570	747K	5.52M	142s

TAKEAWAY

Region count alone is not a useful predictor of spatial workload.

INTERPRETATION

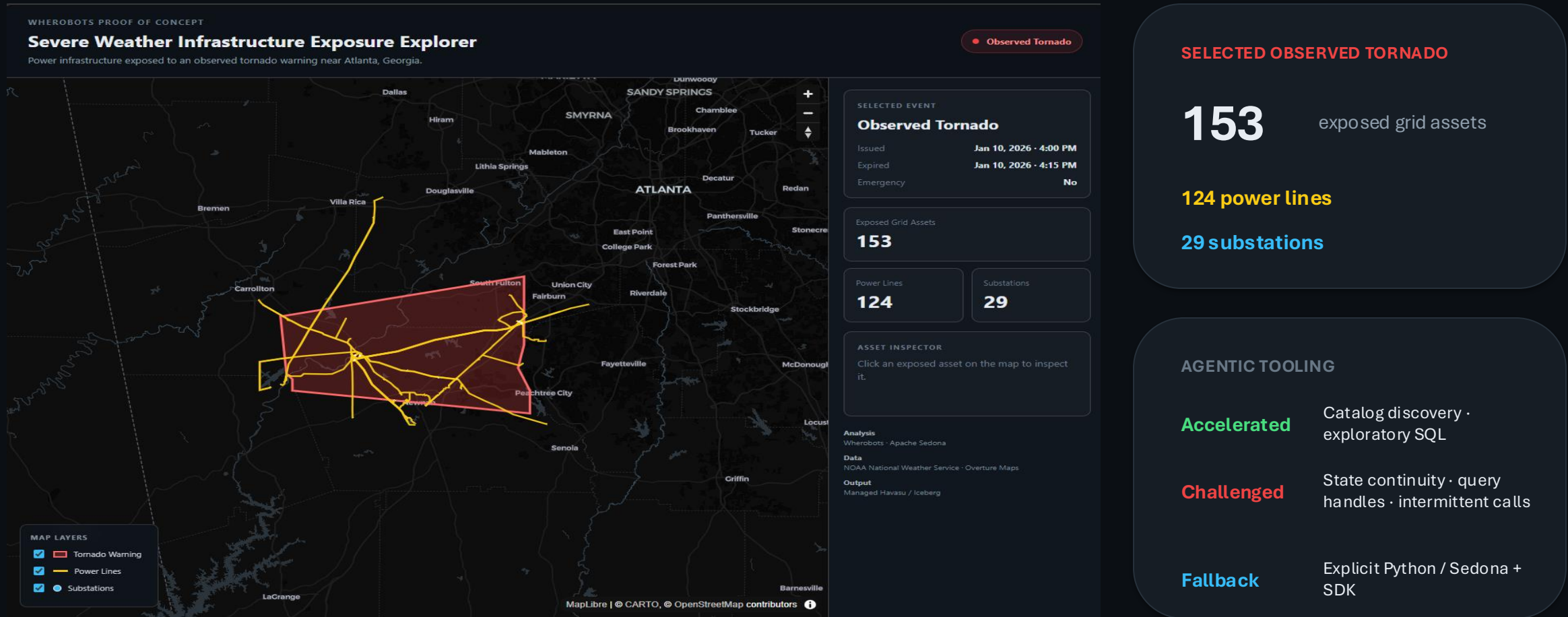
Runtime was not monotonic. These are scale explorations - not a deterministic benchmark - and the Small runtime never hit a demonstrated resource wall.

AT 10-100×

Partition geographically / temporally · inspect spatial partitioning & shuffle · test power_pole · consider incremental processing · resize only after finding the resource boundary

From exposure records to operational decision support

Persisted Wherobots output → lightweight customer-facing explorer.



Next customer question: How do you define asset criticality - and how should exposure events enter the operational response workflow?

Recommended next steps

Move from proof of exposure to an operational utility workflow.

01

Validate the operating model

Confirm which assets are truly critical, how operators prioritize response, and where exposure alerts should enter existing workflows.

02

Enrich with utility context

Join customer-owned criticality, redundancy, outage history, service territory, and crew/access data to the exposure records.

03

Operationalize event processing

Move from annual analysis toward incremental warning ingestion, durable history, and the serving/API layer needed by operations.

04

Scale against the real workload

Test denser infrastructure classes and larger territories; inspect spatial joins, partitioning and runtime requirements using measured production patterns.

Decision to make together: What operational action should a severe-weather exposure event trigger?

Know what is exposed. Add the context that determines what matters.

Wherobots turns large-scale spatial data into a repeatable exposure workflow that can become more valuable as utility-specific operational context is added.

Spatial foundation

NOAA + Overture
Apache Sedona

Operational data product

Parameterized job
Managed Havasu / Iceberg

Path to action

Customer criticality
Response workflow

Thank you

Questions & discussion